

Dark Fifth Forces

(aka long-range SIDM/aka VFM)

Center for Computational
Astrophysics

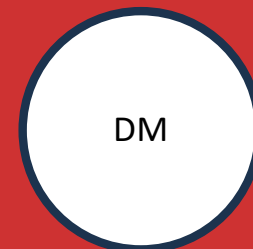
Cosmology Summer Fest 2025



Olivier
Simon

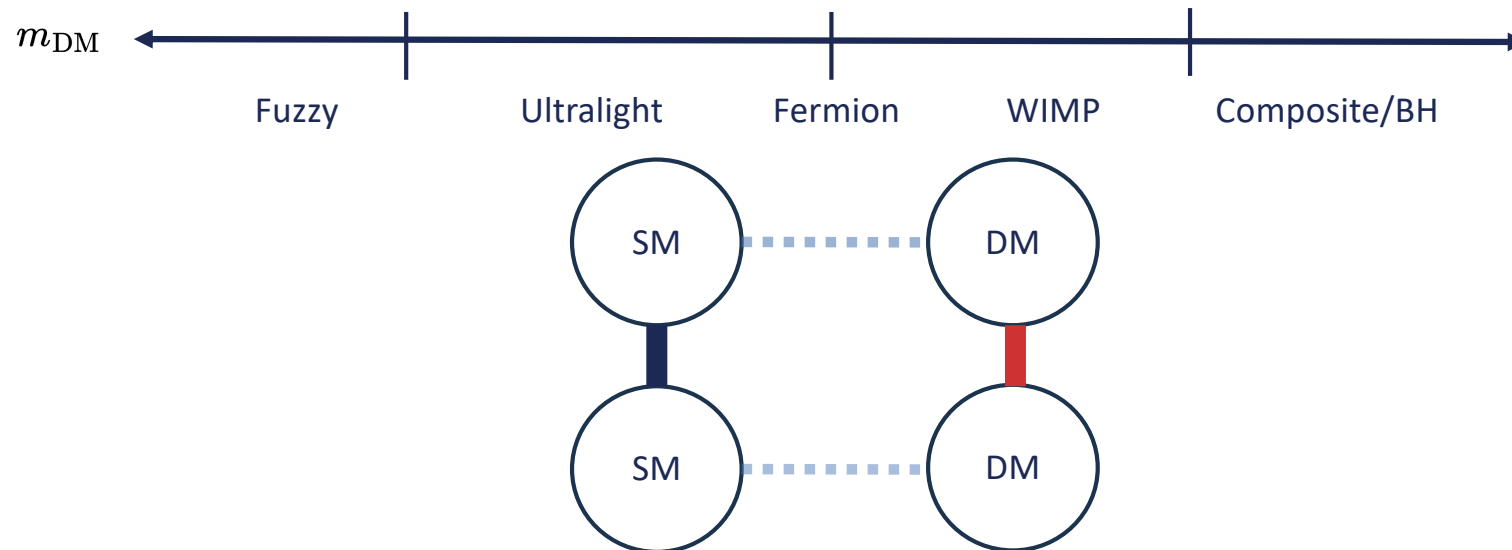


Ken Van
Tilburg



07/23/2025

What is Dark Matter?



Nightmare Scenario

Two Paths in the Nightmare Scenario

“Chameleon”

$$\mathcal{L} = - \left(\frac{1}{2} \partial_\mu \chi \partial^\mu \chi + \frac{1}{2} m_\chi^2 \chi^2 \right) - \left(\frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi + \frac{1}{2} m_\varphi^2 \varphi^2 \right) - A \varphi \chi^2$$

“Symmetron”

$$\mathcal{L}_\pm = - \left(\frac{1}{2} \partial_\mu \chi \partial^\mu \chi + \frac{1}{2} m_\chi^2 \chi^2 \right) - \left(\frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi \pm \frac{1}{2} m_\varphi^2 \varphi^2 \right) \pm \frac{1}{2} g \varphi^2 \chi^2$$

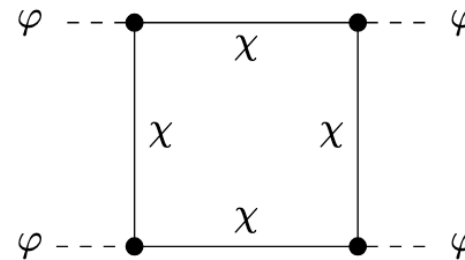
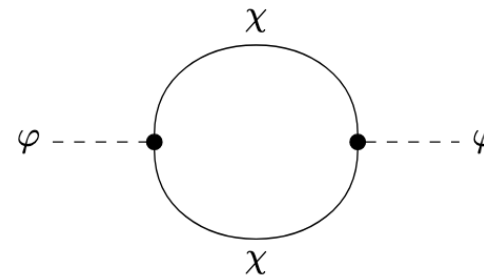
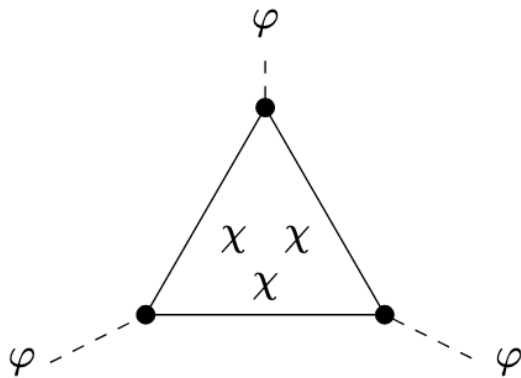
[Kesden & Kamionkowski: astro-ph/0606566, astro-ph/0608095
2204.08484, 2309.11496, 1107.2112, etc ...]

“Chameleon” Dark Fifth Force

$$\mathcal{L} = - \left(\frac{1}{2} \partial_\mu \chi \partial^\mu \chi + \frac{1}{2} m_\chi^2 \chi^2 \right) - \left(\frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi + \frac{1}{2} m_\varphi^2 \varphi^2 \right) - A \varphi \chi^2$$

Technical Naturalness

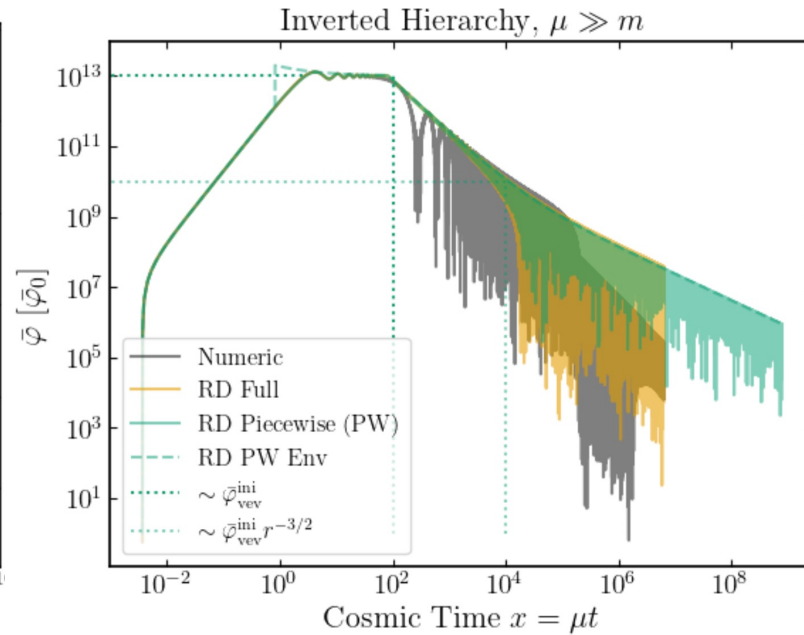
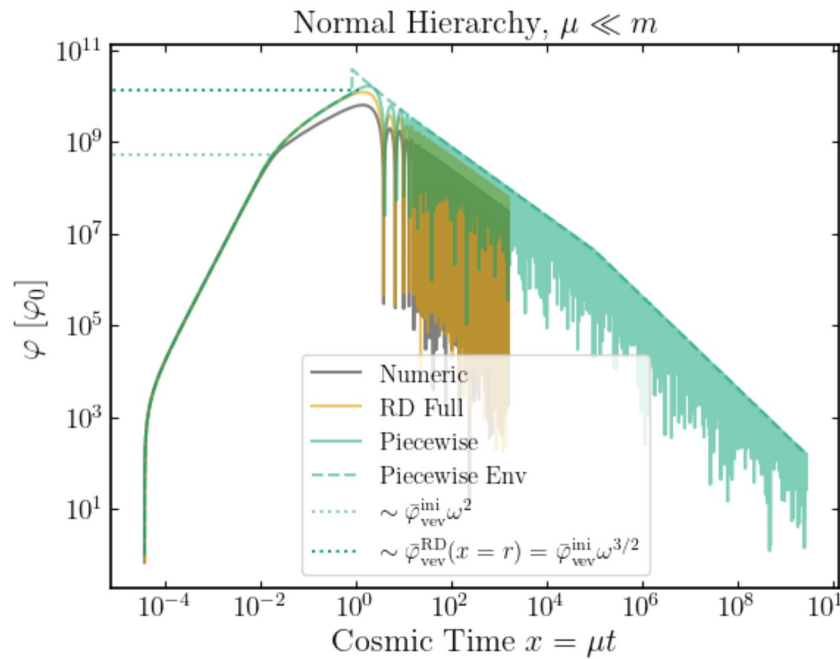
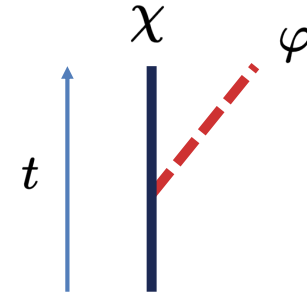
$$m_\varphi^2 \gtrsim \left(\frac{A}{4\pi} \right)^2$$



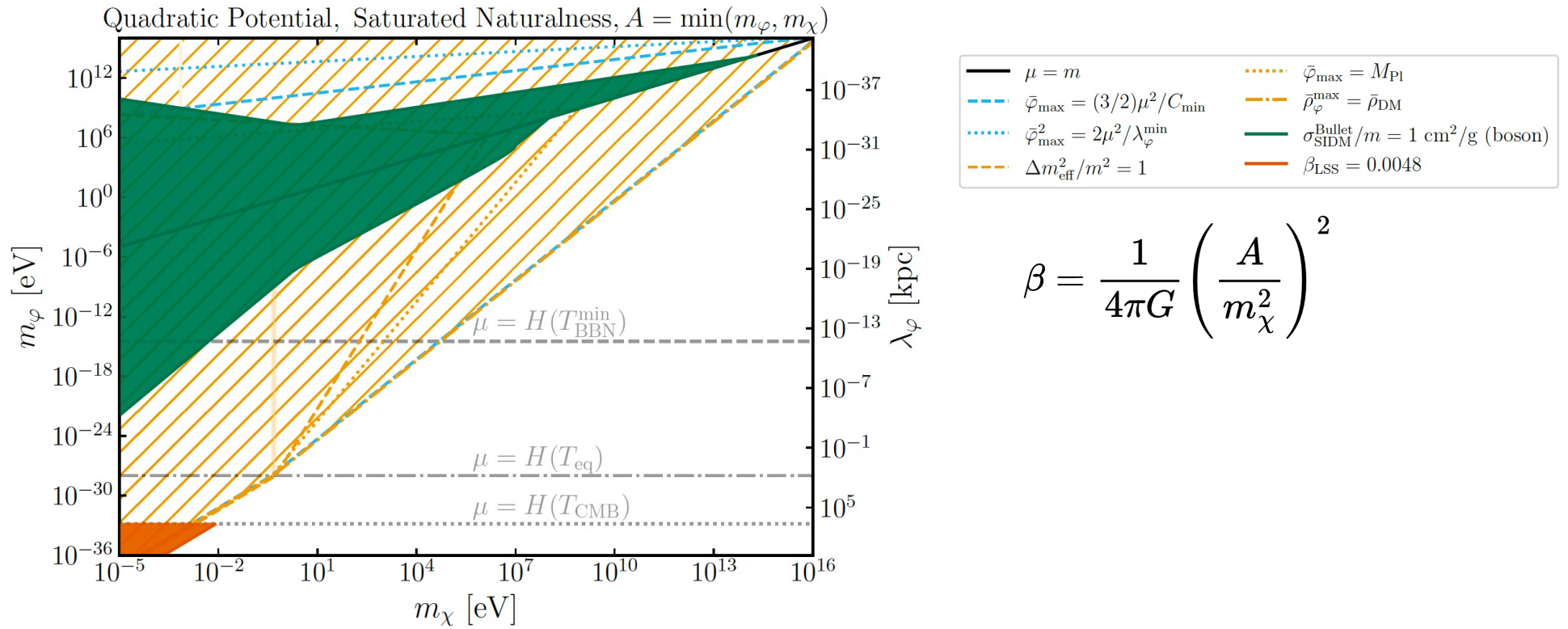
Background Cosmology

$$\ddot{\bar{\varphi}} + 3H\dot{\bar{\varphi}} + m_{\varphi}^2\bar{\varphi} = -A\bar{\chi}^2$$

$$\ddot{\bar{\chi}} + 3H\dot{\bar{\chi}} + (m_{\chi}^2 + 2A\bar{\varphi})\bar{\chi} = 0$$

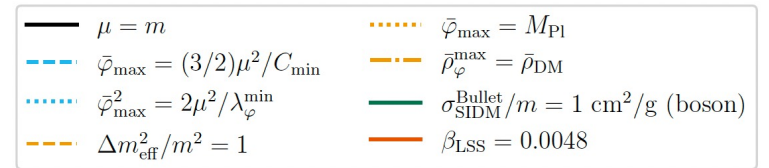
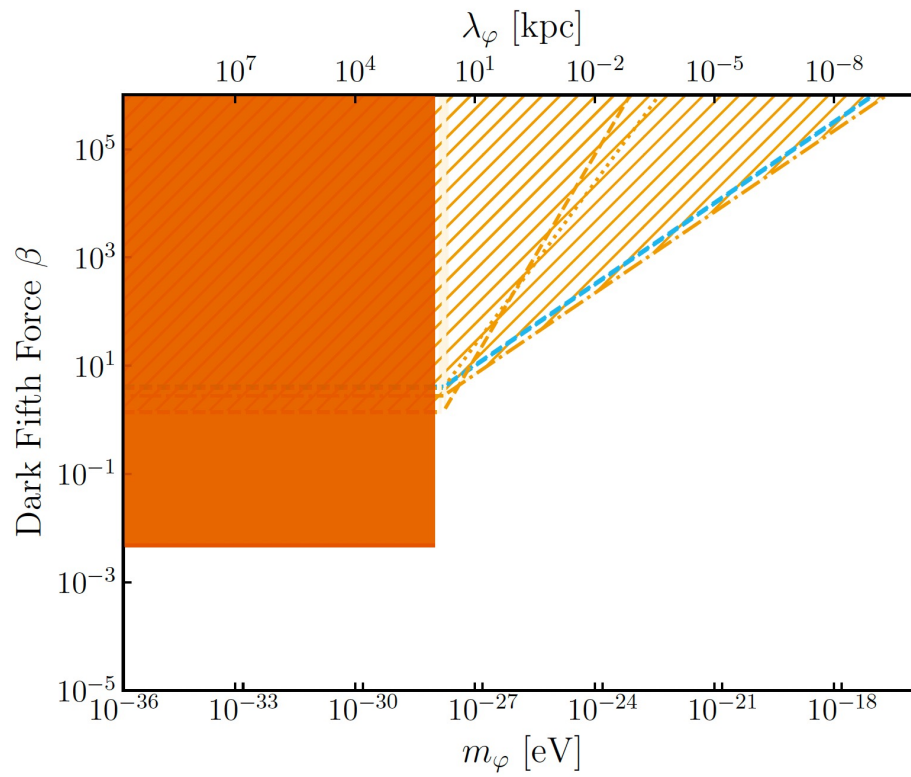


Parameter Space



$$\beta = \frac{1}{4\pi G} \left(\frac{A}{m_\chi^2} \right)^2$$

Parameter Space



Effective Fluids

Timescale Hierarchy $m/H_0 \gg 1$

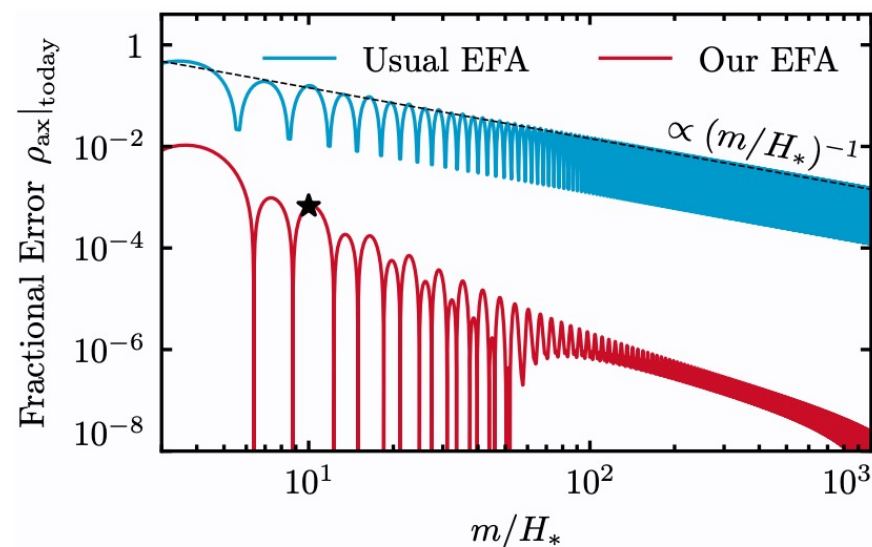


$$\phi(\tau) = \varphi_c(\tau) \cos[\tau - \tau_*] + \varphi_s(\tau) \sin[\tau - \tau_*]$$

$$\rho_{\text{ax}}^{\text{ef}} \equiv \frac{1}{2} m^2 \left(\varphi_c^2 + \varphi_s^2 + \frac{\varphi_c'^2}{2} + \frac{\varphi_s'^2}{2} - \varphi_c \varphi_s' + \varphi_s \varphi_c' \right)$$

$$P_{\text{ax}}^{\text{ef}} \equiv \frac{1}{2} m^2 \left(\frac{\varphi_c'^2}{2} + \frac{\varphi_s'^2}{2} - \varphi_c \varphi_s' + \varphi_s \varphi_c' \right)$$

$$\dot{\rho}_{\text{ax}}^{\text{ef}} = -3 \frac{\dot{a}}{a} (\rho_{\text{ax}}^{\text{ef}} + P_{\text{ax}}^{\text{ef}})$$



[Passaglia & Hu 2201.10238,
Baryakhtar, Simon, Weiner 2405.10358,
Liu, Hu, & Grin 2412.15192]

Cosmological Perturbations

$$\begin{aligned}\delta\ddot{\varphi} + 3H\delta\dot{\varphi} + \left[\frac{k^2}{a^2(t)} + m_\varphi^2 \right] \delta\varphi &= -2A\bar{\chi}\delta\chi - \frac{1}{2}\dot{h}\dot{\varphi} \\ \delta\ddot{\chi} + 3H\delta\dot{\chi} + \left[\frac{k^2}{a^2(t)} + (m_\chi^2 + 2A\bar{\varphi}) \right] \delta\chi &= -2A\bar{\chi}\delta\varphi - \frac{1}{2}\dot{h}\dot{\chi}\end{aligned}$$

How to obtain effective fluid?

Multiple Scale Analysis

Dimensionless Variables

$$x \equiv m_\chi t, y \equiv \varepsilon x, z \equiv \varepsilon^2 x, , \bar{X} \equiv \bar{\chi}/\bar{\chi}_{\text{ini}}^\varphi \text{ and } \bar{\Phi} \equiv \bar{\varphi}/\bar{\chi}_{\text{ini}}^\varphi$$

MSA Form

$$\begin{aligned} (\partial_x + \varepsilon \partial_y + \varepsilon^2 \partial_z)^2 \bar{X} + (1 + 2\alpha \bar{\Phi}) \bar{X} &= 0 \\ (\partial_x + \varepsilon \partial_y + \varepsilon^2 \partial_z)^2 \bar{\Phi} + \mu^2 \bar{\Phi} &= -\alpha \bar{X}^2 \end{aligned}$$

MSA Hierarchy

$$\begin{aligned} \partial_x^2 \bar{\Phi}_0 + \mu^2 \bar{\Phi}_0 &= 0 \\ \partial_x^2 \bar{X}_0 + \bar{X}_0 &= 0 \\ \partial_x^2 \bar{\Phi}_1 + \mu^2 \bar{\Phi}_1 &= -2\partial_x \partial_y \bar{\Phi}_0 - \alpha_1 \bar{X}_0^2 \\ \partial_x^2 \bar{X}_1 + \bar{X}_1 &= -2\partial_x \partial_y \bar{X}_0 - 2\zeta_1 (\bar{\Phi}_1^{\text{vev}} + \bar{\Phi}_0^{\varphi \text{osc}}) \bar{X}_0 - 2\alpha_1^{(2)} \bar{\Phi}_1^{\chi \text{osc}} \bar{X}_0 \end{aligned}$$

Cosmological Perturbations

$$\begin{aligned}\delta\ddot{\varphi} + 3H\delta\dot{\varphi} + \left[\frac{k^2}{a^2(t)} + m_\varphi^2 \right] \delta\varphi &= -2A\bar{\chi}\delta\chi - \frac{1}{2}\dot{h}\dot{\varphi} \\ \delta\ddot{\chi} + 3H\delta\dot{\chi} + \left[\frac{k^2}{a^2(t)} + (m_\chi^2 + 2A\bar{\varphi}) \right] \delta\chi &= -2A\bar{\chi}\delta\varphi - \frac{1}{2}\dot{h}\dot{\chi}\end{aligned}$$

$$c_X^2 \equiv \frac{1}{\omega_\chi^2(t)} \left(\frac{k^2}{4} - \frac{A^2 \bar{n}_\chi}{\omega_\chi(t)(k^2 + m_\phi^2)} \right) \quad c_{\phi_0}^2 \approx \left(1 - \frac{A^2}{m_\phi^2} \frac{\bar{n}_\chi}{m_\chi^3} \right) \frac{k^2}{4m_\phi^2}$$

Conclusion

1. Generically, dark Fifth Forces yield mixed dark sectors
2. Large dark fifth-forces at intermediate scales are currently unconstrained.
3. Multiple Scale Analysis is powerful!
 - Effective Fluid can be split into a “dressed” field and free off-shell mediators
 - Perturbative expansion
4. Input to non-linear simulations (e.g. Isabel Sands,
Olivier Simon, Mariangela Lisanti)
5. Maybe no enhancements? (with Zach Weiner, Marco Costa, Olivier Simon)
6. Spatially-Varying Dark Energy?

Dark matter is elusive but...



Happy Birthday
Marc and Robert!



Technical Naturalness in More Detail

“Chameleon”

$$L \gtrsim A \left(\frac{\Lambda_{\text{UV}}}{4\pi} \right)^2 + C \left(\frac{\Lambda_{\text{UV}}}{4\pi} \right)^2, \quad m_\varphi^2 \gtrsim \left(\frac{A}{4\pi} \right)^2 + \lambda_\varphi \left(\frac{\Lambda_{\text{UV}}}{4\pi} \right)^2, \quad C \gtrsim \frac{A}{(4\pi)^2} \left(\frac{A}{m_\chi} \right)^2, \quad \lambda_\varphi \gtrsim \frac{1}{(4\pi)^2} \left(\frac{A}{m_\chi} \right)^4$$

“Symmetron”

$$m_\chi^2 \gtrsim g \left(\frac{\Lambda_{\text{UV}}}{4\pi} \right)^2, \quad m_\varphi^2 \gtrsim g \left(\frac{\Lambda_{\text{UV}}}{4\pi} \right)^2, \quad \lambda_\varphi \gtrsim \frac{g^2}{(4\pi)^2},$$